MUHAMMAD HUZAIFA RAHIM MALIK

Electronics Engineer | Robotics, Autonomous Systems & Embedded Hardware

Personal Details

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Profile

Electronics engineer with 4+ years of experience in hardware development, embedded systems, robotics, and applied R&D. Experienced in designing, integrating, and field-testing autonomous UGVs and multi-sensor platforms, custom PCBs, embedded AI computing and SDR systems. Combines system-level architecture with sensor integration, SLAM/Nav2-based navigation and LLM-driven robot control. Seeking graduate research training in robotics and autonomous systems to bridge research and deployable engineering.

Education and Qualifications

Quaid-i-Azam University (QAU), Islamabad, Pakistan Sept 2017 – Oct 2021
B.S. Electronics | CGPA: 3.7/4.0 | Status: Completed

- **Thesis: Design and Implementation of an IR-Sensor Speed Detection System with Integrated DC Motor Ballistic Deployment**
  Abstract: Designed and implemented an Arduino-based velocity measurement system utilizing infrared sensors to precisely calculate the speed of fast-moving objects, specifically table tennis balls, with real-time LCD output. Engineered a synchronized automated ball launcher powered by DC motors to consistently deploy objects into the sensor surveillance zone, validating system accuracy and responsiveness.

Professional Experience

Assistant Manager (Tech) July 2024 – Present
AITeC, National Centre for Physics (NCP), Islamabad

- Led electrical design and integration of three autonomous platforms (indoor robot, outdoor vehicle, quadruped) in the AAI&DS lab, from concept to field-ready deployment.
- Developed LLM-driven natural-language navigation for the TurtleBot 4 and extended it to multi-robot coordination with object tracking on a custom mini swarm.
- Built a fault-simulation testbed for the Predictive Maintenance lab to collect realistic industrial fault data for AI models.
- Developed autonomous outdoor point-to-point navigation for an X500 UAV.

Junior Design Engineer (Hardware) Apr 2022 – June 2024
BCUBE Pvt. Ltd., Rawalpindi

- Conducted R&D on an upgraded Software-Defined Radio (SDR) platform: design, structural modifications, link-budget analysis, and PCB layout in Altium Designer.
- Designed a 4-layer digital control PCB for an RF signal source, plus the AGC-splitter and variable-gain pre-amplifier boards of an AGC receiver chain.
- Defined and ran test procedures from bare PCB to system level, including SDR debugging, loom verification, and integration testing of a 12-module VPX-based control unit.

Research Officer (Internee) Nov 2021 – Jan 2022
Pakistan Detector Technologies Pvt. Ltd., Islamabad

- Conducted R&D on metal detector systems; contributed to design, simulation, and verification of detector circuits.

Research and Projects

Autonomous Artificial Intelligence and Decision Support (AAI&DS) Lab, AITeC

- **LLM-based navigation on TurtleBot 4:** Built a natural-language navigation system integrating multiple Whisper speech-recognition and Ollama LLM models (selectable at runtime) to turn voice and chat commands into navigation goals executed through Nav2/SLAM, with a GUI providing control buttons and sensor visualization.
- **Multi-robot navigation and object tracking:** Extended the TurtleBot 4 navigation system with camera-based object tracking and a custom-designed mini swarm vehicle, enabling leader-follower multi-robot coordination in an indoor environment.
- **Outdoor autonomous vehicle (AT-IntelliRover):** Designed the electrical architecture of a 4WD ROS 2 robot built around an NVIDIA AGX Orin and Orin Nano Super, with a low-level motor controller, 3D LiDAR, multi-camera, GPS, and high-torque encoder motors, plus a 4-layer custom PCB power distribution and protection board for the vehicle designed in Altium, which interconnected all the modules.
- **Indoor autonomous robot:** Architected the full electrical system and sensor suite (camera, IMU, 2D LiDAR, encoders) around an NVIDIA Jetson Nano, with Arduino-based PID motor control and Altium-designed schematics, for real-time navigation and algorithm development.
- **Quadruped robot:** Designed the electronic architecture and schematics for a 12-DOF quadruped robot, incorporating a Raspberry Pi 4B and an I2C servo controller. Developed a custom three-rail buck power system (dual 6.5V/15A servo rails, single 5V Pi rail) based on worst-case load-current analysis, powered by an 11.1V LiPo supply with integrated fan cooling.
- **Drone navigation and obstacle avoidance:** Implemented autonomous coordinate-based outdoor navigation on an X500 V2 platform with a Pixhawk 6X flight controller. Tuned system parameters and integrated RC transmitter switches for real-time mission configuration

Predictive Maintenance and Inference Lab (PA&I), AITeC

- **Industrial test-rig fault simulator:** Designed Altium power distribution circuits and sensor interfaces for a rotary fault simulator powered by 24/12/5V ATX supplies. Integrated MPU9250 accelerometers, current, and level sensors into an NI USB-6008 DAQ system for real-time data acquisition.
- **Synthetic dataset generation:** Developed a LabVIEW acquisition pipeline synchronized with Arduino-streamed accelerometer data (Python), logging timestamped multi-sensor runs across no-load and 2/5/7/10 kg load cases for AI-based predictive analytics.

Hardware Design and Test, BCUBE Pvt. Ltd.

- **RF signal-source digital control (4-layer PCB):** Designed the digital section of an ADF4350-based board, integrating an ATmega328PB microcontroller, FT232RL USB-to-UART interface, and a level translator to program the PLL synthesizer and control power-rail enables and RF switching.
- **AGC-Splitter board:** Designed an EMI/EMC-filtered ±15 V supply, cascaded TL061 inverting op-amp gain stages with trimmer-adjustable gain and offset, THS3121 current-feedback output buffers, and a comparator/MOSFET diode-switch test stage.
- **Variable-gain pre-amplifier board:** Designed around an AD605 AGC amplifier, with 15 V→5 V→1.25 V buck regulation, a transformer-based signal-addition stage, and an AD817 unity-gain buffer (±34 dB gain range).
- **VPX-based control unit:** Performed bring-up, debugging, and integration testing of a 12-module unit with custom NVIDIA-based compute cards, three multi-band RF front ends, per-band baseband cards, power supplies, and an external adapter.

Academic Projects, Quaid-i-Azam University

- **Object speed detector:** Arduino-based system using IR sensors to measure and display object speed on an LCD, with a DC motor-driven ball thrower for controlled testing.
- **Image defogging:** Applied image-processing techniques in Jupyter Notebook to remove fog from images, evaluating output quality using signal-to-noise ratio (SNR).

Research Interests

Autonomous robotics (UGV/UAV), SLAM and navigation, LLM-enabled human-robot interaction, embedded AI systems, hardware design and system integration.

Publications

1. M. H. R. Malik, M. A. Khan, M. Rizwan, M. M. Razzaqi, and M. A. Kamran, "A Hybrid Multimodal LLM Navigation Technique for Indoor Ground Mobile Robots: A Fully Local Implementation on TurtleBot 4," submitted to *International Conference on Computing, Intelligence and Innovation (ICCII 2026)*, Oct. 2026. — Under Review

2. M. Rizwan, M. A. Khan, M. H. R. Malik, M. M. Razzaqi, and M. A. Kamran, "Marker-Free Infrastructure-Based Localization of Multiple Ground Robots with Two Stereo Depth Cameras," submitted to *8th International Conference on Advancements in Computational Sciences (ICACS'27)*, Sep. 2026. — Under Review

Skills

Hardware & PCB:	Altium Designer, Multisim, Proteus
Embedded & Robotics:	Arduino, ESP32, Raspberry Pi, Jetson Orin Nano, AGX Orin; LiDAR, IMU, GPS, camera and encoder integration; Nav2, SLAM
Programming:	C++, Python, Git
Tools & OS:	PyCharm, VS Code, Jupyter Notebook, LabVIEW; Ubuntu, Windows

Training and Workshops

- NAIS 2025 (conference participant)
- Vibration Analysis; AI-Enabling Predictive Maintenance (technical training)

Interests and Extra-Curricular Activities

- **Sports:** Cricket, Football, Table Tennis
- **Hobbies:** Travelling, Watching Movies
- **Languages:** English, Urdu, Punjabi